

1. Administrative & Study Identification

Full Study Title: A Phase III Multi-center, Randomized, Open-label Trial to Evaluate Efficacy and Safety of Ribociclib With Endocrine Therapy as an Adjuvant Treatment in Patients With Hormone Receptor-positive, HER2-negative Early Breast Cancer
Short Title/Acronym: NATALEE Study (New Adjuvant TriAl With Ribociclib [LEE011]: NATALEE) **Protocol Number:** CLEE011012301C (STUDY00143618) **NCT Number:** NCT03701334 **Posting ID:** 662779187042283955 **Sponsor Name:** Novartis **Investigational Product:** Ribociclib (Trade Name: Kisqali; ATC Code: L01EF02) + Endocrine Therapy (ET: Letrozole, Anastrozole, Goserelin) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of ribociclib in combination with Endocrine Therapy (ET) versus ET alone in improving invasive disease-free survival (iDFS) as an adjuvant treatment in women and men with HR+, HER2- Early Breast Cancer (EBC). **Secondary Objectives:** To evaluate recurrence-free survival (RFS), distant disease-free survival (DDFS), overall survival (OS), patient-reported outcomes, pharmacokinetics, and safety and tolerability.

2.2 Background & Rationale To address the risk of disease recurrence in patients with HR+/HER2- stage II and III early breast cancer, including those with node-negative disease. Compared with the 600-mg/day dose used in advanced breast cancer, the reduced ribociclib dose (400 mg/day) used in NATALEE aims to improve tolerability while maintaining efficacy over a prolonged 36-month adjuvant treatment duration, maximizing the benefit of CDK4/6 inhibition.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Endocrine Therapy alone) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: 5,101 patients
Trial Status: Active, not recruiting (ACTIVE_NOT_RECRUITING)
Number of Sites: Multicenter (Global) **Estimated Study Start:** 24-Jun-2019 **Estimated Primary Completion:** Q4-2025 (Long-term follow-up ongoing)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of informed consent signature. 2. Female with known menopausal status at the time of randomization or initiation of adjuvant ET (whichever occurs earlier), or male. 3. Histologically confirmed unilateral primary invasive adenocarcinoma of the breast with a date of initial cytologic or histologic diagnosis within 18 months prior to randomization. 4. Confirmed diagnosis of HR-positive (ER and/or PgR positive), HER2-negative Early Breast Cancer. 5. Tumor was completely removed by surgical resection, with final surgical specimen microscopic margins free from tumor, belonging to anatomic stage group II or III. 6. Available archival tumor tissue from the surgical specimen. 7. Completed adjuvant and/or neoadjuvant chemotherapy or radiotherapy (if indicated) per institutional guidelines. 8. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 9. No contraindication for adjuvant ET and planned to be treated with ET for 5 years.

4.2 Exclusion Criteria 1. Distant breast cancer metastasis beyond regional lymph nodes (Stage IV according to AJCC 8th edition) and/or evidence of recurrence after curative surgery. 2. Prior treatment with any CDK4/6 inhibitor. 3. Prior treatment with tamoxifen, raloxifene, or AIs for reduction in risk ("chemoprevention") of breast cancer and/or treatment for osteoporosis within the last 2 years prior to randomization, or concurrent use of hormone replacement therapy. 4. Prior treatment with anthracyclines at cumulative doses of 450 mg/m² or more for doxorubicin, or 900 mg/m² or more for epirubicin. 5. Known hypersensitivity to any of the excipients of ribociclib and/or ET. 6. Concurrent use of other anti-neoplastic therapy (except adjuvant ET) or major surgery, chemotherapy, or radiotherapy within 14 days prior to randomization. 7. Unrecovered clinical and laboratory acute toxicities related to prior anti-cancer therapies. 8. Concurrent invasive malignancy or a prior invasive malignancy whose treatment was completed within 2 years before randomization. 9. Known HIV infection, Hepatitis B, or C infection. 10. Clinically significant, uncontrolled heart disease and/or cardiac repolarization abnormality. 11. Current use (within 7 days before randomization) of strong CYP3A4/5 inhibitors/inducers, or medications with a narrow therapeutic window predominantly metabolized through CYP3A4/5. 12. Current or prior use of systemic corticosteroids ≤ 2 weeks prior to starting trial treatment. 13. Impairment of GI function or GI disease that may significantly alter absorption of oral trial treatments. 14. Participation in other studies involving investigational drug(s) within 30 days prior to randomization or 5 half-lives. 15. Pregnant or breast-feeding (lactating) women or women planning to become pregnant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ribociclib 400 mg by mouth once daily on days 1 to 21 of a 28-day cycle, for 36 months (approximately 39 cycles). Endocrine therapy consists of letrozole 2.5 mg or anastrozole 1 mg daily (with goserelin 3.6 mg SC every 4 weeks for men and premenopausal women). **Route of Administration:** Oral (Ribociclib, Letrozole, Anastrozole) and Subcutaneous (Goserelin). **Duration of Treatment:** 36 months of Ribociclib; 60+ months of Endocrine Therapy.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications per local clinical guidelines. **Prohibited:** Concurrent use of strong CYP3A4 inhibitors/inducers, other experimental anti-cancer agents, or additional systemic targeted therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Screening, Informed Consent, Medical History, Tumor and Blood Samples
2	0	Baseline, Randomization, First Dose
3	1 of Cycles	Interim, Safety Labs (CBC, LFTs, ECGs), Efficacy Assessment, ET Compliance, PROs
4	36 / 60+	End of Study, Final Vital Signs, iDFS/OS Tracking, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Invasive Disease-Free Survival (iDFS) based on STEEP criteria (Standardized Definitions for Efficacy End Points in Adjuvant Breast Cancer Trials), as assessed by the Investigator. iDFS is defined as the time from the date of randomization to the date of the first event of local invasive breast recurrence, regional invasive recurrence, distant recurrence, death (any cause), contralateral invasive BC, or second primary non-breast invasive cancer (excluding basal and squamous cell carcinomas of the skin).

7.2 Secondary Endpoints 1. Recurrence-free survival (RFS) 2. Distant disease-free survival (DDFS) 3. Overall survival (OS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient interviews, physical exams, and central/local laboratory evaluations (specifically monitoring for neutropenia, QT prolongation, and hepatotoxicity). **Serious Adverse Events (SAEs):** Required to be reported to the sponsor and regulatory authorities within 24 hours of site awareness. **Data Monitoring Committee**

(DMC): Independent Data Monitoring Committee (IDMC) performing regular safety and efficacy interim analyses.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes (iDFS/OS). Safety Set for all patients receiving at least one dose of study drug. **Sample Size Justification:** Powered to detect a statistically significant improvement in iDFS for the ribociclib + ET arm versus ET alone using a stratified log-rank test. **Handling of Missing Data:** Standard Kaplan-Meier estimates with event censoring at the date of the last adequate disease assessment for patients without an iDFS event.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol compliance and patient safety. **Audit Trail:** Validated eCRF tracking, electronic patient diaries for PROs, and continuous source data verification (SDV).

11. Study Identifiers & Governance

Primary ID: CLEE011012301C **Registry Number:** NCT03701334 **Posting ID:** 662779187042283955 **Sponsor/Lead Organization:** Novartis **Study Title:** NATALEE **EMA Regulatory Information:** [Kisqali EPAR Product Information](#)

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Early Breast Cancer (EBC) **Disease Definition:** Early Breast Cancer refers to a stage of breast cancer where the malignancy is confined to the breast or has spread only to regional lymph nodes, most commonly in the axilla, without evidence of distant metastasis to other organs. This stage typically involves smaller tumors and is generally associated with a more favorable prognosis and higher rates of successful treatment. **Diagnosis Threshold:** HR-positive, HER2-negative, Anatomic Stage Group II or III **Preferred Nomenclature:** BREAST CANCER, EARLY-ONSET; Early Hormone Receptor Positive HER2 Negative Breast Cancer **Ontology Code:** SNOMED ID: 109869006 (Hierarchy: 254837009 | Malignant neoplasm of breast | Malignant neoplasm of breast (disorder)) **Medical Methodology:** Adjuvant Endocrine Therapy (ET) plus CDK4/6 Inhibition **Optimization**

Target: Extension of Invasive Disease-Free Survival (iDFS) and prevention of recurrence

13. Study Procedures & Methodology

Management Pathway: Standard adjuvant clinical pathway for localized, high-risk early breast cancer. **Education Protocol (HCP):** Site initiation visits covering the specialized 400 mg ribociclib dose modification scheme and safety management (ECG and liver function monitoring). **Education Protocol (Patient):** Dosing diaries (3 weeks on/1 week off schedule) and side-effect reporting education. **Quality Audit Mechanism:** Routine cross-referencing of pharmacy dispensing logs with patient electronic compliance diaries.

14. Observation & Follow-Up Schedule

Total Duration: Ribociclib treatment for 36 months; Endocrine Therapy for 60 months; long-term survival follow-up. **Onsite Visit Cadence:** Every cycle (28 days) during the active 36-month Ribociclib treatment phase. **Remote Monitoring Frequency:** Intermittent survival and disease status checks during the long-term follow-up phase. **Checklist Review Interval:** Symptom checklist and adverse event review at Day 1 of every cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				
Statistician	[Name]	_____	[DD-MMM-YYYY]				